

Neurodevelopmental Disorder with Microcephaly, Seizures, and Cortical Atrophy (NDMSCA / VARS1-related disorder)

Comprehensive Disease Characteristics Report OMIM #617802 · Gene: VARS1 · Category: Mendelian (autosomal recessive) *Report compiled from a 5-iteration autonomous literature investigation (30 papers reviewed, 9 confirmed findings)*

Summary

Neurodevelopmental disorder with microcephaly, seizures, and cortical atrophy (NDMSCA; also abbreviated NEDMISCA), OMIM #617802, is an ultra-rare autosomal recessive Mendelian disease caused by biallelic variants in *VARS1*, the gene encoding the sole cytoplasmic valyl-tRNA synthetase (ValRS). The disease was established in 2019 by two independent landmark cohorts — Siekierska et al. (10 patients; [PMID: 30755616](#)) and Friedman et al. (7 patients from 5 families; [PMID: 30755602](#)) — and was substantially expanded in 2026 by Aynekin et al. (13 individuals from 10 families; [PMID: 41672381](#)). In total, roughly ~30 **molecularly confirmed patients** are described in the world literature.

Affected individuals present with a consistent core phenotype: **primary or progressive microcephaly, global developmental delay / intellectual disability, early-onset epilepsy or epileptic encephalopathy, and progressive cerebral cortical atrophy with white-matter volume loss**. The molecular mechanism is a **partial loss of valyl-tRNA aminoacylation function** — most pathogenic alleles are hypomorphic missense variants clustered in the tRNA-binding/anticodon and catalytic domains — which impairs cytoplasmic protein translation in the developing brain. Functional confirmation comes from reduced enzymatic activity in patient fibroblasts, yeast complementation assays, and a *vars*-knockout zebrafish model that recapitulates key neurological traits.

There is **no disease-specific or curative therapy**; management is entirely supportive (antiseizure medications, developmental/rehabilitative therapy, feeding support). Mechanistically rational but unproven experimental strategies — cognate amino acid (valine) supplementation, cognate-tRNA delivery, and integrated-stress-response (ISR) inhibition — are borrowed from the broader aminoacyl-tRNA synthetase (ARS) disorder field. Prognosis is poor, with severe lifelong disability, frequently drug-resistant epilepsy, and a neurodegenerative course; residual VARS1 enzymatic activity of the specific genotype is the most plausible prognostic modifier.

1. Disease Information

Overview. NDMSCA is a Mendelian, autosomal recessive, infantile-onset neurodevelopmental and neurodegenerative disorder of cytoplasmic protein translation. It belongs to the growing class of **aminoacyl-tRNA synthetase (ARS) deficiencies**, in which impaired charging of a specific tRNA with its cognate amino acid disrupts protein synthesis, with a brain-predominant clinical picture.

Key identifiers:

Resource	Identifier
OMIM (phenotype)	#617802 — "Neurodevelopmental disorder with microcephaly, seizures, and cortical atrophy"
OMIM (gene)	192150 (VARS1)
Gene symbol	VARS1 (formerly VARS)
HGNC	HGNC:12651
NCBI Gene	7407
Ensembl	ENSG00000204394
UniProt	P26640
MONDO (suggested)	MONDO:0060639
Cytogenetic locus	6p21.33 (within the MHC region), chr6:31,777,518–31,795,752 (GRCh38)

Synonyms / alternative names: NDMSCA; NEDMISCA; VARS1-related (valyl-tRNA synthetase) developmental encephalopathy; "developmental encephalopathy with microcephaly"; progressive neurodevelopmental epileptic encephalopathy due to VARS variants.

Source of information. Evidence is derived from **aggregated disease-level and primary case-series resources** (OMIM, published patient cohorts), not from EHR/individual-patient population databases. The disease-level knowledge base comprises three cohort publications plus isolated case reports.

2. Etiology

Primary cause — genetic. The disease is caused exclusively by **biallelic (homozygous or compound heterozygous) pathogenic variants in VARS1**. As Friedman et al. state: "*VARS encodes the only known valine cytoplasmic-localized aminoacyl-tRNA synthetase. Here, we report seven patients from five unrelated families with five different biallelic missense variants in VARS*" ([PMID: 30755602](#)). There is no environmental or infectious contribution to primary causation.

Genetic risk factors. The only established risk factor is inheritance of two damaging *VARS1* alleles. **Consanguinity** increases risk, as expected for a recessive disorder — homozygous variants are reported in multiple families, including consanguineous unions (e.g., the consanguineous case in [PMID: 36204440](#)).

Environmental risk / protective factors. None identified. No lifestyle, occupational, toxic, or nutritional factors are known to modify onset or risk. As a fully penetrant monogenic recessive disorder, gene–environment interactions have not been described. There are no reported genetic protective/modifier alleles beyond the intrinsic effect of residual enzyme activity (see §9, §11).

3. Phenotypes

The core phenotype is a **triad of progressive microcephaly, seizures, and cortical atrophy**, on a background of global developmental delay / intellectual disability. Friedman et al.: "*Subjects present with a range of global developmental delay, epileptic encephalopathy and primary or progressive microcephaly. Longitudinal assessment demonstrates progressive cortical*

atrophy and white matter volume loss" (PMID: 30755602). Aynekin et al. confirm the triad in an independent cohort: "a neurodevelopmental syndrome with progressive microcephaly, seizures, and intellectual disability" (PMID: 41672381).

Phenotype	Type	Onset	Severity / course	Frequency (qualitative)	Suggested HPO
Microcephaly (primary or progressive)	Physical/ growth sign	Congenital–infantile	Moderate–severe; often progressive	Core (near-universal)	HP:0000252; HP:0000253 (progressive)
Global developmental delay / intellectual disability	Cognitive/ behavioral	Infantile	Severe	Core	HP:0001263; HP:0001249
Seizures / epileptic encephalopathy	Neurological sign	Early-onset (often infantile)	Often drug-resistant; severe	Common/ core	HP:0001250; HP:0200134
Cerebral cortical atrophy	Imaging sign	Infantile onset, progressive	Progressive	Core (imaging)	HP:0002120
White-matter volume loss / abnormal white matter	Imaging sign	Progressive	Progressive	Common	HP:0002500
Hypotonia	Neurological sign	Infantile	Variable	Common	HP:0001252
Thin corpus callosum, delayed myelination	Imaging sign	Infantile	Structural	Variable/ common	HP:0002079; HP:0012448
Spasticity, feeding difficulties, visual/optic involvement	Neurological/ other	Variable	Variable	Variable	(feature-dependent)

Quality-of-life impact. Given severe intellectual disability, drug-resistant epilepsy, motor impairment, and feeding difficulty, affected children are typically fully dependent for daily activities. Formal QoL instrument data (EQ-5D, SF-36, PROMIS) are **not available** for this ultra-rare cohort.

4. Genetic / Molecular Information

Causal gene. *VAR1* (HGNC:12651; OMIM gene 192150; UniProt P26640), encoding cytoplasmic valyl-tRNA synthetase, the enzyme that charges tRNA-Val with valine for cytoplasmic translation.

Pathogenic variants. The mutational spectrum is dominated by **missense variants**, with a minority of truncating alleles:

- Siekierska et al.: an allelic series of **9 biallelic variants (7 novel, 2 previously reported) in 10 patients** (PMID: 30755616).
- Friedman et al.: **5 biallelic missense variants** across 5 families; variants "*map to the VAR1 tRNA binding domain and adjacent to the anticodon domain, and disrupt highly conserved residues*" (PMID: 30755602).
- Aynekin et al. (2026): **15 variants** — 2 causing premature truncation (LOF) and 13 missense localizing to catalytic and aminoacylation domains (PMID: 41672381).

Variant classification / functional consequence. Most alleles are ACMG **pathogenic/likely pathogenic**, but VUS are common across ARS genes (over 80% of ARS missense variants are VUS per PMID: 42028791), underscoring the value of functional assays. The functional consequence is **partial loss of function (hypomorphic)** — patient cells retain intact VAR1 protein but show reduced enzymatic activity (see §6). Complete biallelic null is likely incompatible with life (consistent with embryonic lethality of ARS knockouts in mouse).

Population allele frequency / constraint. gnomAD (GRCh38) constraint for *VAR1*: **pLI $\approx$ 0 (3.8×10^{-15})**, observed/expected LoF = 0.57 (LOEUF $\approx$ 0.68; 92 observed vs 161 expected LoF), **missense Z = 3.71** (oe_mis = 0.79). Interpretation: heterozygous LoF is *tolerated* (carriers unaffected — as expected for a recessive gene), while *strong missense constraint* indicates missense changes in the aminoacylation/tRNA-binding domains are deleterious — matching the observation that most pathogenic alleles are missense.

Somatic vs germline. All variants are **germline**. No somatic involvement.

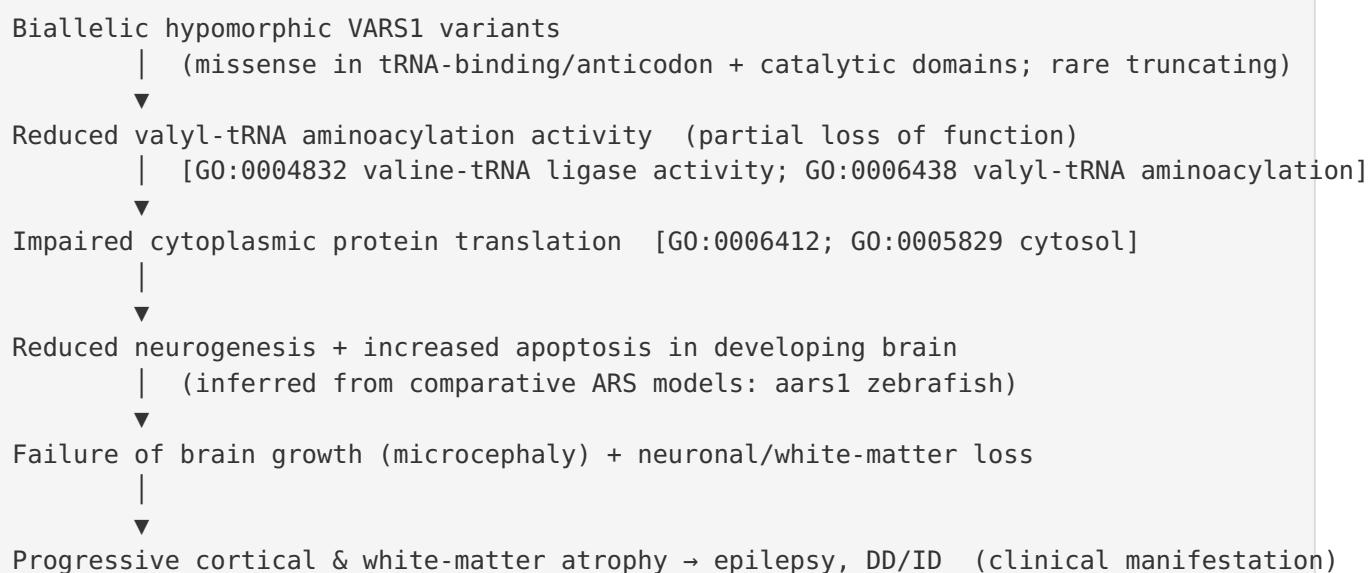
Modifier genes / epigenetics / chromosomal abnormalities. No modifier genes, epigenetic marks, or large-scale chromosomal abnormalities have been described for this disorder. The most likely intrinsic modifier of severity is the **residual aminoacylation activity of the specific biallelic genotype**.

5. Environmental Information

Not applicable. NDMSCA is a purely Mendelian genetic disorder. No environmental factors, lifestyle factors, or infectious agents contribute to causation or triggering. (Note: because *VAR1* lies within the 6p21.33 MHC region, the gene is physically near immune loci, but there is no immune-mediated or infectious mechanism to the disease.)

6. Mechanism / Pathophysiology

Causal chain



Molecular pathway / biochemical defect. The primary lesion is **enzymatic**: reduced valine-tRNA ligase activity (GO:0004832) impairing valyl-tRNA aminoacylation (GO:0006438), a core step of cytoplasmic translation (GO:0006412). Friedman et al.: "Patient primary cells show intact

VARs protein but reduced enzymatic activity, suggesting partial loss of function" (PMID: 30755602). Siekierska et al. independently confirm: "In silico, in vitro, and yeast complementation assays demonstrate that the underlying pathomechanism of these mutations is most likely a loss of protein function" (PMID: 30755616).

Structural basis. Molecular-dynamics simulations show missense substitutions *"can disrupt local protein dynamics, RNA-interaction surfaces, or catalytic geometry, thereby affecting ligand recognition, substrate specificity, and tRNA interaction"* (PMID: 41672381).

Cellular processes. By analogy with the related recessive ARS disorder caused by *AARS1*, the cellular consequence of translation-factor ARS LOF is **reduced neurogenesis and increased apoptosis**: *"zebrafish mutants in aars1 have reduced neurogenesis and increased apoptosis"* (PMID: 41508610). A plausible downstream contributor across ARS diseases is chronic activation of the **integrated stress response (ISR)** secondary to accumulation of uncharged tRNA (the rationale for ISR-inhibitor therapy; PMID: 39702998).

Cell types / subcellular localization. Implicated cell types: cortical neurons (CL:0000540 neuron; CL:0010012 cerebral cortex neuron) and — via white-matter loss/hypomyelination — oligodendrocytes (CL:0000128). *VARs1* acts in the **cytoplasm/cytosol (GO:0005829)**, the site of cytoplasmic translation. (*VARs1* is distinct from mitochondrial ValRS; the disorder is a *cytoplasmic* translation defect.)

Immune / metabolic / omics. No autoimmune or immunodeficiency component. No specific metabolic, proteomic, metabolomic, or lipidomic disease signature has been published beyond the direct aminoacylation defect. No single-cell/spatial/multi-omics datasets exist for this disorder.

7. Anatomical Structures Affected

Organ / body system. The disease is **CNS-predominant**; the primary affected organ is the **brain** (UBERON:0000955). There is no consistent primary involvement of non-neural organs despite *VARs1* being a general translation enzyme.

Regional / tissue involvement (bilateral, diffuse):

Structure	UBERON	Finding
Cerebral cortex	UBERON:0000956	Progressive cortical atrophy
Cerebral white matter	UBERON:0002316	Progressive white-matter volume loss; delayed myelination
Corpus callosum	UBERON:0001851	Thinning
Brain (global)	UBERON:0000955	Microcephaly / reduced brain volume

Friedman et al.: *"Longitudinal assessment demonstrates progressive cortical atrophy and white matter volume loss"* (PMID: 30755602).

Cell level: cortical neurons (CL:0000540, CL:0010012); oligodendrocytes (CL:0000128). **Subcellular:** cytosol (GO:0005829). **Lateralization:** bilateral/symmetric, diffuse.

8. Temporal Development

Onset. Congenital-to-early-infantile. Microcephaly may be **primary (present at birth)** or **progressive (postnatally acquired)**; epilepsy is often early-onset. Siekierska et al.: *"ten patients with a developmental encephalopathy with microcephaly, often associated with early-onset epilepsy"* (PMID: 30755616).

Progression. The course is **progressive / neurodegenerative** — longitudinal imaging demonstrates worsening cortical and white-matter atrophy over time (PMID: 30755602). Friedman et al. explicitly frame VARS1 disease as *"pediatric neurodegeneration."* Disease duration is chronic and lifelong; there is no remission.

Critical period. Because the defect impairs neurogenesis during brain development, the **prenatal and early-postnatal window** is the period of greatest vulnerability and the theoretical window for any disease-modifying intervention.

9. Inheritance and Population

Epidemiology. Ultra-rare. The world literature comprises approximately ~30 **molecularly confirmed individuals**: 10 (Siekierska et al., [PMID: 30755616](#)), 7 from 5 families (Friedman et al., [PMID: 30755602](#)), 13 from 10 families (Aynekin et al., [PMID: 41672381](#)), plus isolated reports (e.g., [PMID: 36204440](#)). **No population prevalence or incidence estimate exists** (not established in Orphanet/GBD).

Inheritance. Autosomal recessive. Affected individuals carry biallelic (homozygous or compound heterozygous) *VAR1* variants; parents are unaffected obligate heterozygotes — consistent with the LoF-tolerant heterozygous gnomAD profile. Aynekin et al.: *"We clinically evaluated 13 affected individuals from 10 unrelated families"* ([PMID: 41672381](#)); Friedman et al.: *"seven patients from five unrelated families with five different biallelic missense variants"* ([PMID: 30755602](#)).

- **Penetrance:** Presumed complete for biallelic damaging genotypes.
 - **Expressivity:** Variable (severity likely tracks residual enzyme activity; Aynekin et al. define new clinical/molecular subtypes).
 - **Consanguinity:** Increases risk; homozygous cases reported in consanguineous families.
 - **Founder effect / anticipation / mosaicism:** No established founder mutation; two recurrent alleles noted by Siekierska et al. suggest limited allelic recurrence. No anticipation (not a repeat-expansion disorder). No germline mosaicism reported.
 - **Carrier frequency:** Not formally established; very low given rarity.
 - **Sex ratio:** No sex bias (autosomal). Both sexes affected.
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10. Diagnostics

Molecular diagnosis is definitive. All reported patients were identified by **whole-exome or whole-genome sequencing** revealing biallelic *VAR1* variants ([PMID: 30755616](#); [PMID: 30755602](#); [PMID: 41672381](#)). WES/WGS and neurodevelopmental/epilepsy gene panels including *VAR1* are the recommended first-line tests. Chromosomal microarray/karyotype are not informative (no structural cause).

Functional confirmation. A **patient-fibroblast aminoacylation activity assay** confirms pathogenicity and reclassifies VUS. A high-throughput LC-MS/MS aminoacylation assay measuring aaRS activity in patient fibroblasts *"has contributed to the diagnosis of nearly 200 patients"* across ARS genes (PMID: 42028791).

Supportive investigations. Serial **brain MRI** (NCIT:C16809) shows progressive cortical atrophy, white-matter volume loss, thin corpus callosum, delayed myelination; **EEG** (NCIT:C38054) documents epileptic encephalopathy. Head-circumference tracking documents microcephaly.

Suggested NCIT terms: Whole Exome Sequencing (NCIT:C101294), MRI (NCIT:C16809), EEG (NCIT:C38054).

Differential diagnosis. Other ARS/translation disorders and recessive microcephaly–epilepsy–cortical-atrophy syndromes reviewed during this investigation, distinguished by gene and neuroimaging pattern: *AARS1*, *EPRS1* (HLD15), *DARS2/AARS2/EARS2* (mitochondrial leukoencephalopathies), *ASNS* deficiency, *TBCD* tubulinopathy, *TRAPPC4*, *BRAT1*, *CSTB*, *MINPP1*, *UFM1* (H-ABC), *WWOX*. Genetic testing is required to distinguish these overlapping phenotypes.

Screening. No newborn/population screening exists. Cascade carrier testing of relatives and prenatal/preimplantation testing are feasible once the familial variants are known.

11. Outcome / Prognosis

Prognosis is poor. Affected individuals have **profound intellectual and motor disability with frequently drug-resistant epilepsy**, on a **progressive/neurodegenerative** trajectory. Friedman et al. frame the disorder as pediatric neurodegeneration: *"The implication of VARS in pediatric neurodegeneration broadens the spectrum of human diseases due to mutations in tRNA synthetase genes"* (PMID: 30755602).

Survival / mortality. Formal survival, mortality, and QoL statistics are **not available** for this ~30-patient cohort. Early-childhood death occurs at the severe end of the spectrum (as seen across severe recessive translation disorders).

Prognostic modifier. The most plausible predictor of severity is the **residual VARS1 aminoacylation activity** of the specific biallelic genotype — truncating/low-activity alleles predicting greater severity — consistent with the partial-loss-of-function model and the new molecular subtypes defined by Aynekin et al. (PMID: 41672381).

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive and symptomatic**:

Domain	Intervention	Suggested NCIT
Epilepsy	Antiseizure medications (often multiple; may be refractory)	NCIT:C264 (Anticonvulsant Agent)
Development	Physical/occupational/speech therapy	—
Nutrition	Feeding support (may require tube feeding)	—
Monitoring	Serial MRI, EEG, growth tracking	NCIT:C16809, NCIT:C38054

Experimental / mechanism-based strategies (unproven for VARS1):

- **Cognate amino acid (valine) supplementation** — CHEBI:16414 (valine). *"Current treatment approaches to rescue defective or dysfunctional tRNA synthetase mutants include supplementation with cognate amino acids and delivery of cognate tRNAs to alleviate bottlenecks in translation. Complementary approaches use inhibitors to target the integrated stress response"* (PMID: 39702998).
- **Cognate-tRNA delivery and ISR inhibitors** — investigational across ARS disorders (PMID: 39702998; PMID: 42028791).

Important caveat on efficacy. Amino acid supplementation is **safe but of unproven and apparently limited efficacy**. In a pilot trial of related mitochondrial ARS leukoencephalopathies (AARS2/DARS2), supplementation was *"safe and well tolerated ... but efficacy endpoints were not met"* though most patients remained clinically stable (PMID: 41075682). Anecdotal benefit has been reported in some other ARS disorders (e.g., methionine in MARS1, tyrosine in YARS2), highlighting variable, gene- and case-specific responses. No controlled trial exists specifically for VARS1/valine.

13. Prevention

There is **no primary prevention** for this genetic disorder beyond reproductive genetic counseling.

- **Genetic counseling:** Autosomal recessive recurrence risk of 25% for future pregnancies of carrier couples; counseling is the central preventive tool.
- **Carrier / cascade testing:** Once familial *VARs1* variants are known, cascade testing of relatives and reproductive partners is possible.
- **Prenatal / preimplantation genetic testing (PGT):** Available for at-risk couples with known variants.
- **Consanguinity awareness:** In consanguineous populations, awareness and pre-conception counseling reduce recurrence.

Immunization, behavioral, environmental, and public-health prevention are **not applicable**.

14. Other Species / Natural Disease

- **Orthologous genes / taxonomy:** zebrafish *vars/vars1* (*Danio rerio*, NCBI Taxon 7955); *Saccharomyces cerevisiae* ortholog *VAS1* (Taxon 4932). *VARs/ValRS* is deeply evolutionarily conserved as an essential translation enzyme.
 - **Natural disease in other species:** No naturally occurring *VARs1* disorder in companion animals or wildlife has been reported (not applicable per OMIA).
 - **Comparative biology:** The disease mechanism (essential aminoacylation) is conserved across eukaryotes, which is why yeast complementation is a valid pathogenicity assay.
 - **Zoonotic potential:** Not applicable (non-infectious).
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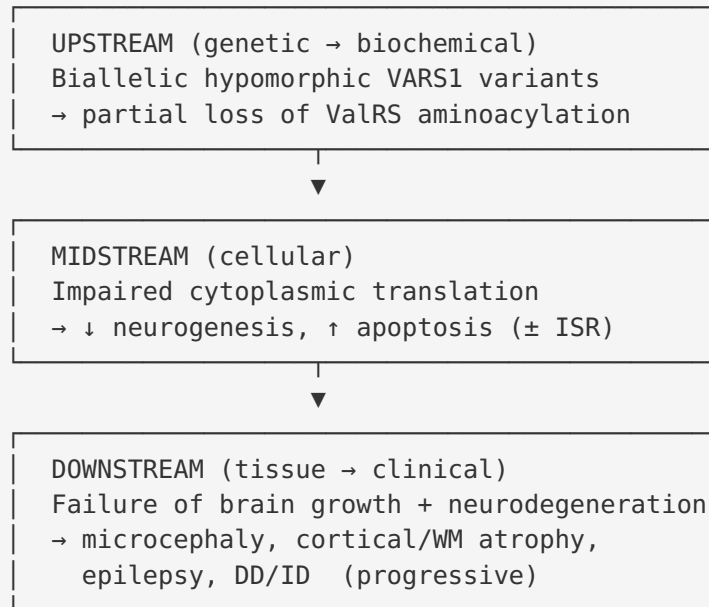
15. Model Organisms

Model	Type	Key finding	Resource
<i>vars</i> -knockout zebrafish	Vertebrate (whole-organism)	"Zebrafish modeling accurately recapitulated some of the key neurological disease traits" (PMID: 30755616)	ZFIN (Taxon 7955)
Yeast complementation (<i>S. cerevisiae</i>)	Cellular / functional	Demonstrated loss of function of patient variants (PMID: 30755616)	SGD (Taxon 4932)
Patient fibroblasts	In vitro (human)	Intact protein but reduced aminoacylation activity (PMID: 30755602)	—
Comparative: <i>aars1</i> zebrafish / mouse	Vertebrate	"zebrafish mutants in <i>aars1</i> have reduced neurogenesis and increased apoptosis"; Aars1 mouse embryonic-lethal (PMID: 41508610)	ZFIN, MGI

Phenotype recapitulation. The zebrafish *vars* knockout reproduces key neurological traits, validating it as a disease model; yeast complementation validates individual variant pathogenicity. **Limitations:** neither model fully captures the human progressive cortical/white-matter atrophy or seizure phenotype in detail. A dedicated *Vars1* mouse (knock-in of hypomorphic alleles) would better model the partial-LOF neurodegenerative course — this is a knowledge gap.

Mechanistic Model / Interpretation

The evidence converges on a **single coherent mechanism** unified across three independent human cohorts and multiple functional systems:



Three features make this model robust: (1) **convergent human genetics** — biallelic *VARS1* in ≥ 30 patients across ≥ 17 families from independent groups; (2) **direct functional evidence** — reduced enzyme activity in patient cells with intact protein (a *hypomorphic*, not null, mechanism); and (3) **cross-system validation** — yeast complementation and zebrafish knockout. The hypomorphic nature explains why the disease is viable at all (complete null likely lethal) and predicts a **genotype–severity relationship driven by residual activity** — the single most important open translational question.

Evidence Base

PMID	Role	How it supports the findings
30755602	Landmark cohort (Friedman)	Establishes biallelic missense <i>VAR1</i> cause, partial-LOF mechanism (reduced enzyme activity, intact protein), progressive cortical/WM atrophy, "pediatric neurodegeneration."
30755616	Landmark cohort (Siekierska)	Allelic series of 9 biallelic variants in 10 patients; yeast + zebrafish confirm loss of function; early-onset epilepsy + microcephaly.
41672381	2026 expansion (Aynekin)	Largest cohort (13/10 families); defines clinical/molecular subtypes; MD simulations of missense variants; truncating + missense spectrum.
42028791	Methods/therapy review	Aminoacylation functional assay for diagnosis/VUS reclassification; amino acid supplementation and gene-therapy landscape for ARS diseases.
39702998	Therapy review	Mechanistic experimental strategies: cognate amino acid supplementation, tRNA delivery, ISR inhibition.
41508610	Comparative model	<i>aars1</i> zebrafish show reduced neurogenesis + increased apoptosis — the cellular basis of ARS microcephaly.
36204440	Case report	VAR1 VUS in a consanguineous neonate with NDMSCA-consistent neuroimaging; illustrates consanguinity and dual-diagnosis complexity.
41075682	Trial (related ARS)	Amino acid supplementation safe but efficacy endpoints not met in AARS2/DARS2 — tempers therapeutic expectations.

Contextual differential-diagnosis literature reviewed (not causal for this disease): MINPP1-PCH (PMID: [41025723](#)), UFM1 H-ABC (PMID: [35189806](#)), ASNS deficiency (PMID: [30978478](#)), TBCD tubulinopathy (PMID: [27807845](#)), TRAPPC4 (PMID: [31794024](#)), CSTB (PMID: [28378817](#)), BRAT1 (PMID: [28635423](#)), EPRS1/HLD15 (PMID: [41721156](#)), MARS1 (PMID: [32833345](#)).

Limitations and Knowledge Gaps

1. **Ultra-rare cohort (~30 patients).** No population prevalence/incidence, no formal survival/mortality/QoL data, and limited natural-history granularity.
 2. **No genotype–phenotype quantification.** Although residual enzyme activity is the presumed severity determinant, no systematic correlation between measured aminoacylation activity and clinical severity has been published.
 3. **No dedicated mammalian (mouse) model** of hypomorphic *Vars1*; existing models (zebrafish KO, yeast) do not fully recapitulate progressive cortical atrophy or epilepsy.
 4. **No omics profiling** (transcriptomic, proteomic, metabolomic) of patient neural tissue or organoids.
 5. **No VARS1-specific therapeutic data.** Valine supplementation is mechanistically rational but untested; extrapolation from other ARS trials suggests safety but uncertain/limited benefit.
 6. **Frequencies of individual features** (e.g., exact % with seizures, hypotonia, feeding difficulty) are not precisely tabulated across the pooled cohort.
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Proposed Follow-up Experiments / Actions

1. **Establish a genotype–activity–severity map.** Systematically measure fibroblast valyl-tRNA aminoacylation activity for each biallelic genotype and correlate with a standardized severity score (seizure control, developmental quotient, rate of atrophy on serial MRI).
2. **Generate a hypomorphic *Vars1* knock-in mouse** (patient missense allele in trans with a null) to model the progressive neurodegenerative course and test interventions in a mammalian brain.
3. **Patient-iPSC cortical organoids** to test whether (a) valine supplementation, (b) cognate-tRNA delivery, or (c) ISR inhibitors (e.g., ISRIB) rescue neurogenesis/apoptosis phenotypes in vitro.
4. **Pilot valine-supplementation study** with pre-specified biomarker endpoints (aminoacylation activity, ISR markers) and imaging endpoints, learning from the null-efficacy AARS2/DARS2 trial design ([PMID: 41075682](#)).
5. **Build an international patient registry** to capture natural history, survival, seizure semiology/EEG evolution, and QoL — prerequisites for any future trial.

6. **Reclassify VUS at scale** using the high-throughput LC-MS/MS aminoacylation assay ([PMID: 42028791](#)) to improve diagnostic yield and enable cascade testing.

Ontology Term Appendix

- **MONDO:** MONDO:0060639 (NDMSCA)
 - **Gene/HGNC:** HGNC:12651 (VARs1); UniProt P26640; NCBI Gene 7407
 - **HPO:** HP:0000252, HP:0000253, HP:0001263, HP:0001249, HP:0001250, HP:0200134, HP:0002120, HP:0002500, HP:0001252, HP:0002079, HP:0012448
 - **GO (process/function/component):** GO:0006438 (valyl-tRNA aminoacylation), GO:0004832 (valine-tRNA ligase activity), GO:0006412 (translation), GO:0000049 (tRNA binding), GO:0005829 (cytosol)
 - **CL:** CL:0000540 (neuron), CL:0010012 (cerebral cortex neuron), CL:0000128 (oligodendrocyte)
 - **UBERON:** UBERON:0000955 (brain), UBERON:0000956 (cerebral cortex), UBERON:0002316 (white matter), UBERON:0001851 (corpus callosum)
 - **CHEBI:** CHEBI:16414 (valine)
 - **NCIT:** NCIT:C101294 (WES), NCIT:C16809 (MRI), NCIT:C38054 (EEG), NCIT:C264 (Anticonvulsant Agent)
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Evidence source types: human clinical/genetic (Siekierska, Friedman, Aynekin cohorts; case reports), in vitro (patient fibroblast aminoacylation assays), model organism (zebrafish, yeast), and computational (molecular dynamics, gnomAD constraint). All mechanistic and clinical claims are cited to primary literature by PMID.
